In – House NTCC Report

On

Satellite Image Analysis

Submitted to

Amity University, Uttar Pradesh

In partial fulfilment of the requirements for the award of the degree of

Bachelor of Technology

In

Computer Science and Engineering

By

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July 2025

DECLARATION

We, **Sneha Tanwar, Aman Gupta** students of B. Tech CSE (5CSE2) hereby declare that the project titled "**Satellite Image Analysis**" which is submitted by us to the **Department of Computer Science and Engineering**, Amity School of Engineering and Technology, Amity University Noida, Uttar Pradesh, in partial fulfilment of requirement for the award of the degree of **Bachelor of Technology in Computer Science and Engineering**, has not been previously formed the basis for the award of any degree, diploma or other similar title or recognition.

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CERTIFICATE

On the basis of the report submitted by **Sneha Tanwar**, **Aman Gupta** students of **B.Tech. (5CSE2)**, I hereby certify that the report entitled "**Satellite Image Analysis**" which is submitted to the **Department of Computer Science and Technology**, Amity School of Engineering and Technology, Amity University Uttar Pradesh in partial fulfilment of the requirement for the award of the degree of **Bachelor of Technology (CSE)** is an original contribution with existing knowledge and faithful record of work carried out by her under my guidance and supervision.

To the best of my knowledge, this work has not been submitted in part or full for any Degree or Diploma to this University or elsewhere.

Noida

Date: Signature of IFC

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ABSTRACT

The accurate localization of typhoons in satellite imagery plays a vital role in meteorological analysis, early warning systems, and disaster response planning. Traditional methods of identifying typhoon centres and cloud structures in satellite images are often time-consuming and rely heavily on manual interpretation, which may lead to inconsistencies and delays. In this study, we propose a deep learning-based approach for precise localization of typhoon features using the YOLOv5 (You Only Look Once, version 5) object detection model. YOLOv5 is well-suited for real-time object detection due to its high inference speed and accuracy, making it an ideal candidate for processing large volumes of satellite data.

A dataset comprising infrared satellite images containing various typhoon events was constructed for training and evaluation. The model was trained to detect typhoon eye regions and dense cloud bands. The model's performance was evaluated using standard object detection metrics, achieving a mean Average Precision (mAP) of 95% at an Intersection over Union (IoU) threshold of 0.5. The results demonstrate the model's robustness in accurately localizing typhoon structures across different image resolutions and typhoon intensities.

This research underscores the effectiveness of YOLOv5 in meteorological applications, particularly for rapid and consistent localization of storm systems in satellite imagery. By enhancing the precision and efficiency of typhoon localization, this method can support meteorologists and disaster management agencies in improving situational awareness and response times during severe weather events.

Keywords – Satellite Imagery, Yolov5, Deep Learning, Image Localization, Typhoon Detection

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CHAPTER 1

INTRODUCTION

A tropical cyclone is a weather phenomenon that consists of whirling thunderstorms and clouds over tropical or subtropical oceans. It has a well-defined centre and constrained low-level circulation, and it is recognized for its tremendous destructive capabilities. These powerful storm systems can generate winds exceeding 74 mph (119 km/h), heavy rainfall, storm surges, and widespread flooding, making them one of nature's most formidable forces.

A tropical depression originates when a closed isobaric circulation develops during the tropical cyclone formation process. As this system strengthens, it may progress to become a tropical storm and potentially a hurricane or typhoon, depending on its location and intensity. Thus, minimizing the hazards demands a dependable system of tropical cyclone identification. Early detection and accurate tracking are crucial for issuing timely warnings to coastal communities and implementing effective evacuation procedures to protect lives and property [1].

To extract useful information from extensive remote sensing data, satellite image processing is necessary. It makes it feasible to observe the Earth's surface and atmosphere over time and space, providing information that is not achievable with only ground-based techniques. Researchers can track weather trends, identify environmental shifts, and investigate dynamic phenomena including cloud movement, land use, sea surface temperatures, and natural disasters by using satellite pictures.

Particularly, infrared satellite imagery can be used to identify cloud formations and temperature changes. This makes it possible to observe continuously and improves our capacity to identify trends, abnormalities, and dynamic systems at high temporal frequencies. Satellite imaging has become a potent tool in a variety of applications, such as meteorology, climate science, environmental monitoring, and disaster response, due to its capacity to automate and scale such study using computer methods. In the end, satellite image analysis aids in converting complicated data into insightful knowledge that promotes scientific comprehension and well-informed decision-making [1], [2].

Components of Interpreting Satellite Images are:

1. Shape: Shape describes an object's geometric form or outlines as seen in satellite images. Regular, well-defined shapes are frequently seen in man-made constructions, such as roads' straight lines or buildings' rectangles. On the other hand, natural features, such as meandering rivers or uneven forest borders, typically have asymmetrical or organic shapes.

2. Size: Size refers to a feature's real dimensions, which need to be assessed considering the image's size and spatial resolution. It aids in distinguishing visually comparable characteristics. It is necessary to comprehend both relative and absolute measurements to evaluate sizes accurately.

3. Pattern: The spatial arrangement and recurrence of characteristics are referred to as patterns. Human activity is frequently indicated by regular and ordered patterns, such as apartment blocks, crop rows, or streets that are precisely aligned.

4. Tone: The brightness or darkness of the image's characteristics, which represent the quantity of electromagnetic energy reflected or emitted, is referred to as tone. The tonal characteristics of various surfaces and materials vary. To differentiate between different types of land cover and surface conditions, tone is an essential component.

One of the most important aspects of catastrophe risk reduction is early tropical cyclone detection. Authorities can drastically lower the number of fatalities and financial loss by issuing early warnings, starting evacuation procedures, and mobilizing emergency resources when identification is done accurately and promptly. Such early warnings are crucial in coastal areas due to the high human density and high susceptibility of infrastructure. The need for reliable and adaptable monitoring systems has increased as cyclone frequency and intensity have increased due to climate change. In addition to facilitating prompt disaster response, early identification improves long-term resilience by guiding emergency preparedness and infrastructure development [2].

A variety of satellite imaging modalities are employed in cyclone detection and monitoring, each offering distinct observational advantages. Infrared (IR) imagery, by measuring thermal radiation, enables continuous monitoring of cloud-top temperatures, which are indicative of storm intensity. Visible imagery provides high-resolution visual representation during daylight hours, useful for assessing cloud structure and eye formation. Water vapor imagery captures upper-tropospheric moisture, offering insights into atmospheric circulation and cyclone development. Microwave sensors penetrate cloud layers, providing data on precipitation and surface wind speed. The synergistic use of multi-spectral data enhances the accuracy of cyclone tracking and intensity estimation.

Even with satellite remote sensing's effectiveness, cyclone identification still faces several obstacles. Early-stage features may be obscured by low spatial and temporal resolution, making it more difficult to discern emerging systems from nearby cloud structures. Continuous monitoring may be hampered by inconsistent data availability brought on by orbital gaps, cloud contamination, or atmospheric interference.

Additionally, overlapping visual features might make it difficult to distinguish tropical cyclones from other convective systems. When machine learning models are trained on datasets that are limited in time or location, they also have problems with generalizability. These drawbacks highlight the necessity of higher resolution, larger datasets, and sophisticated algorithms with strong pattern recognition capabilities [3].

The operational use of satellite imagery has been demonstrated in several major cyclonic events. During Cyclone Amphan (2020), satellite data—particularly infrared and microwave observations—facilitated real-time tracking of storm progression and intensity changes. This enabled authorities in India and Bangladesh to execute early evacuation strategies, significantly reducing loss of life. Similarly, Typhoon Haiyan (2013) was closely monitored using geostationary satellites, which provided continuous coverage of its rapid intensification prior to landfall in the Philippines. These case studies exemplify the critical role of satellite-based monitoring in disaster mitigation and highlight the value of integrating remote sensing into national and regional warning systems.

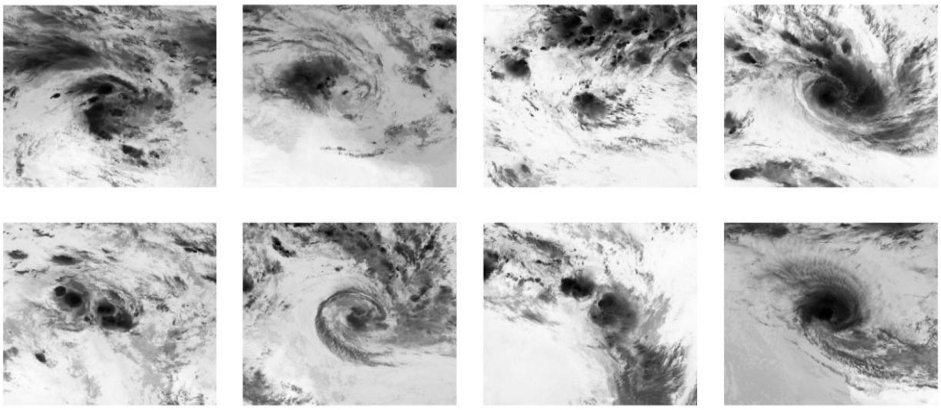


Figure 1: Typhoon Images from the Dataset

CHAPTER 2

PROBLEM STATEMENT

Current typhoon monitoring systems face several operational and analytical challenges that limit their effectiveness during critical response windows. The reliance on manual interpretation of satellite imagery by meteorologists is both time-intensive and prone to subjectivity, often resulting in delays and inconsistent identification of storm characteristics such as spiral bands, eye wall structure, and intensity estimates. Furthermore, the sheer volume of high-frequency satellite data generated daily far exceeds human processing capacity, increasing the likelihood that early-stage or rapidly developing typhoons may go undetected. In many cases, limited spatial and temporal resolution in existing monitoring infrastructure exacerbates these challenges, leaving gaps in coverage that can hinder timely storm tracking. These limitations collectively undermine the accuracy and responsiveness required for effective disaster preparedness and mitigation.

This project proposes a deep learning–based solution to address the limitations of manual typhoon monitoring by introducing an automated framework for satellite image analysis. The primary objective is to develop a model capable of accurately detecting and localizing typhoon formations in infrared satellite imagery, with a focus on minimizing false positives and improving spatial precision. The system is designed to extract meaningful patterns from storm structures and provide consistent, quantitative outputs that can assist meteorologists and emergency response teams in decision-making and situational awareness.

CHATPER 3

LITERATURE REVIEW

Deep learning is a potent approach in image analysis that has made major strides in areas like object detection and semantic segmentation in recent years. Models like UNet, and Transformer have demonstrated strong capabilities in extracting complex features from visual data. These architectures have been widely adopted in remote sensing applications due to their ability to handle high-dimensional, noisy, and multi-scale image features. These segmentation-based approaches, however, have significant drawbacks when used for the job of typhoon centre localization. Typhoons, as mesoscale meteorological phenomena, exhibit irregular shapes, diffuse boundaries, and evolving structures, making pixel-level segmentation methods less effective. These models typically emphasize boundary delineation, whereas typhoon localization requires precise identification of the storm's centre, which is not always visually prominent.

To address these challenges, recent studies have shifted toward object detection techniques, which focus on recognizing and localizing complete targets within an image. Models such as Faster R-CNN, Mask R-CNN, and the YOLO series (including the latest YOLOv11) offer an alternative approach by directly predicting bounding boxes and centre points. These methods have proven more robust in handling the blurred and asymmetric cloud formations typical of typhoons. For instance, some researchers have combined detection networks with meteorological knowledge, such as spiral structures and eye-detection indices, to improve accuracy. Others have enhanced detection performance through hyperparameter tuning or loss function optimization.

Despite these promising developments, most existing research focuses on the application of standard object detection frameworks rather than adapting them to the specific complexities of typhoon localization. To fill this gap, the present study introduces an optimized version of the YOLOv8n model, named TGE-YOLO, which incorporates three key innovations: a TFAM_Concat feature fusion module to enhance focus on typhoon cores, the integration of GSConv to reduce computational cost while preserving feature quality, and an enhanced EIoU loss function to improve localization accuracy by prioritizing centre-point precision. The following sections provide a detailed overview of these advancements in the context of existing literature.

3.1. Existing work on typhoon detection

Early research on typhoon centre localization primarily relied on morphological methods and wind field structure analysis. Hu et al. proposed an azimuthal spectrum analysis approach that identifies the maximum symmetry component surrounding the storm centre to estimate its position.

Zhang et al. introduced a method based on the gradient of brightness temperature and radial symmetry characteristics, applying numerical optimization to enhance localization precision. Similarly, Shin et al. utilized an enhanced logarithmic spiral band (LSB) in conjunction with a scoring matrix (SCM) to detect typhoon centres by capturing spiral structures [4].

While these morphological techniques offer reasonable accuracy under ideal conditions, their effectiveness diminishes in cases where spiral cloud bands or a distinct eyewall are not present. Moreover, rapidly evolving meteorological fields further reduce their reliability. To address these limitations, researchers have explored wind field inversion methods, which localize typhoons by identifying either the zero-wind-speed point or the maximum vorticity within surface wind data. These approaches are particularly helpful during the formation or dissipation stages of a cyclone, especially when visual cues in cloud morphology are unclear. However, their performance declines under intense precipitation or in high-resolution satellite imagery, which introduces noise and ambiguity [5].

With the rise of deep learning, a growing body of research has applied convolutional and transformer-based models to storm analysis. In the field of image segmentation, architectures like UNet and Transformer-based models have shown strong performance in extracting spatial features. The Swin Transformer, introduced by Liu et al., improved representation learning using a hierarchical architecture and sliding window attention, enhancing segmentation outcomes in visual tasks. Further innovations, such as the Vision Mamba UNet (VM-UNet) by Ruan et al, incorporated a Visual State Space (VSS) block to reduce computational complexity while maintaining global contextual awareness. Despite their success in general vision tasks, segmentation models struggle with typhoon localization due to the diffuse boundaries, irregular morphology, and evolving structure of storms. Unlike typical semantic segmentation, where pixel-level accuracy and boundary sharpness are crucial, typhoon localization demands precise centre-point identification, often in scenarios where the centre is visually obscured. This mismatch between task objectives and model architecture limits the applicability of segmentation frameworks for cyclone tracking [6], [7].

In contrast, object detection models have demonstrated greater robustness in this domain, particularly when applied to remote sensing imagery. The models, such as Faster R-CNN, Mask R-CNN, and the

YOLO family, approach localization as a comprehensive object recognition issue by explicitly predicting bounding boxes and object centres [7].

Single-stage detection algorithms like YOLO, due to their real-time capabilities, are particularly well-suited for large-scale satellite data analysis. YOLOv4 has been successfully used by Long et al. to detect typhoon centres from geostationary satellite cloud maps. He et al. applied YOLOX with morphological enhancements to handle high-resolution satellite images containing both typhoon and non-typhoon cloud formations. The latest YOLOv11 introduces architectural improvements (e.g., C3k2 and C2PSA modules) to strengthen feature extraction, though its improvements in detection accuracy remain marginal [4].

Despite the progress made, much of the current literature emphasizes applying existing detection architectures rather than adapting or optimizing them specifically for meteorological applications. Few studies focus on enhancing feature fusion, loss functions, or structural modules tailored to the unique challenges of cyclone detection. This underutilization of algorithmic customization constrains performance, particularly in detecting ambiguous or visually inconsistent typhoon centres.

3.2. YOLO v5

Object identification methods were based on multistage pipelines with independent modules for feature classification, extraction and region proposal. Although effective, these techniques were limited by computational inefficiency and sluggish inference speeds, making them impractical for real-world applications. YOLOv5 is a significant development in real-time object identification that builds on the foundation of the YOLO (You Only Look Once) series. YOLOv5's design is simple and efficient, with a streamlined network consisting of a backbone, neck, and detecting head.

Important elements that make up the YOLOv5 architecture:

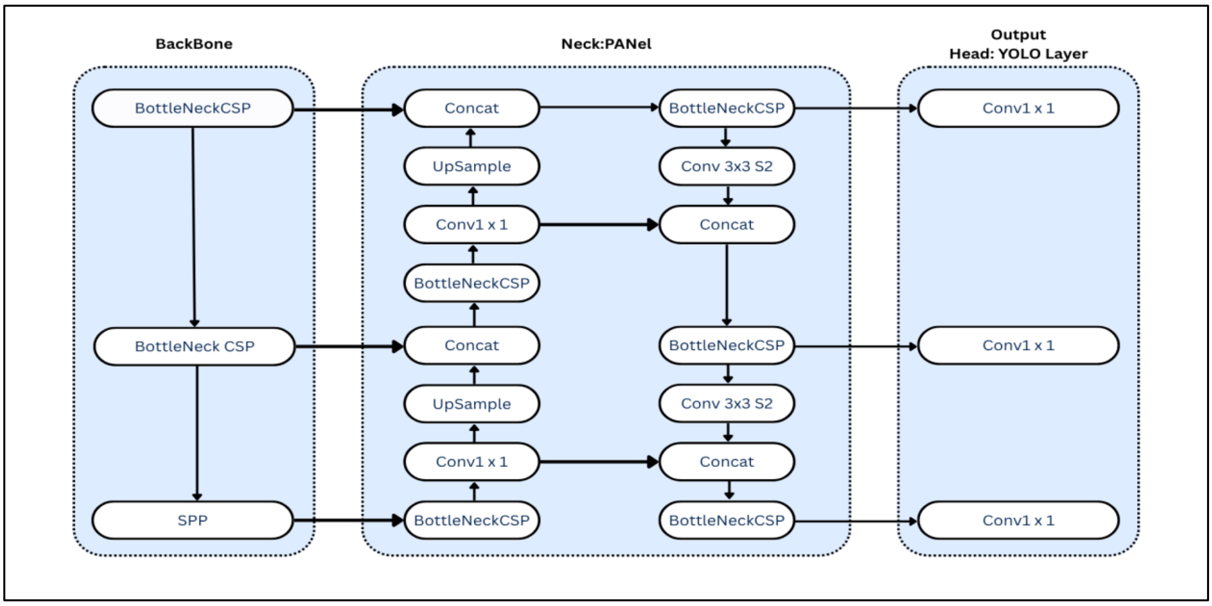


Figure 3.2: YOLOv5 Architecture

1. Backbone Network: YOLOv5 assists a variety of backbone networks, including variants of CSPDarknet, EfficientNet, and ResNet. The backbone network provides rich representations necessary for precise object detection by extracting hierarchical features from incoming photos.
2. Neck Network: It is an intermediate processing step that aggregates and refine features, extracted by the backbone network, which ultimately improves performance.
3. Detection Head: The prediction of bounding boxes, confidence score of each object and their class possibilities are predicted by the detection head of YOLO v5. To produce predictions for object properties, such as object classes and bounding box coordinates, the detecting head combines convolutional layers and activation functions [8].

YOLOv5 was developed as an open-source project by Glenn Jocher et al. From June 2020 to November 2022, it had seven major updates. YOLOv5 is a widely used target detection algorithm that evolved from YOLOv1 over time. It is divided into four models based on the number of parameters: YOLOv5s, YOLOv5m, YOLOv5l, and YOLOv5x. As the number of parameters increases, so does its effectiveness [9]. YOLOv5 includes mosaic data augmentation, which builds on previous data improvement techniques implemented in target detection. The algorithm is enhanced based on the CutMix data enhancement method, which only uses two images for splicing. In contrast, the Mosaic data enhancement technique uses four images and splices them into a single image using random scaling, random cropping, and random arrangement. This makes the dataset richer and significantly increases the network's training speed while lowering the model's memory requirements.

Based on existing data enhancement techniques used for target detection, YOLOv5 incorporates mosaic data augmentation [9].

The Web of Science data indicates that there are more relevant works based on YOLOv5 than on other models in the YOLO series. Because of its good stability and dependability, as well as its ease of deployment and training, YOLOv5 (GitHub-ultralytics/yolov5: YOLOv5) is preferred [10]. In several YOLO models, it is evident that YOLOv5 has high competition. Consequently, the YOLOv5 model was selected as the main application model for the typhoon monitoring studies in this work. One popular deep-learning framework is called YOLOv5.

CHAPTER 4

METHODOLOGY

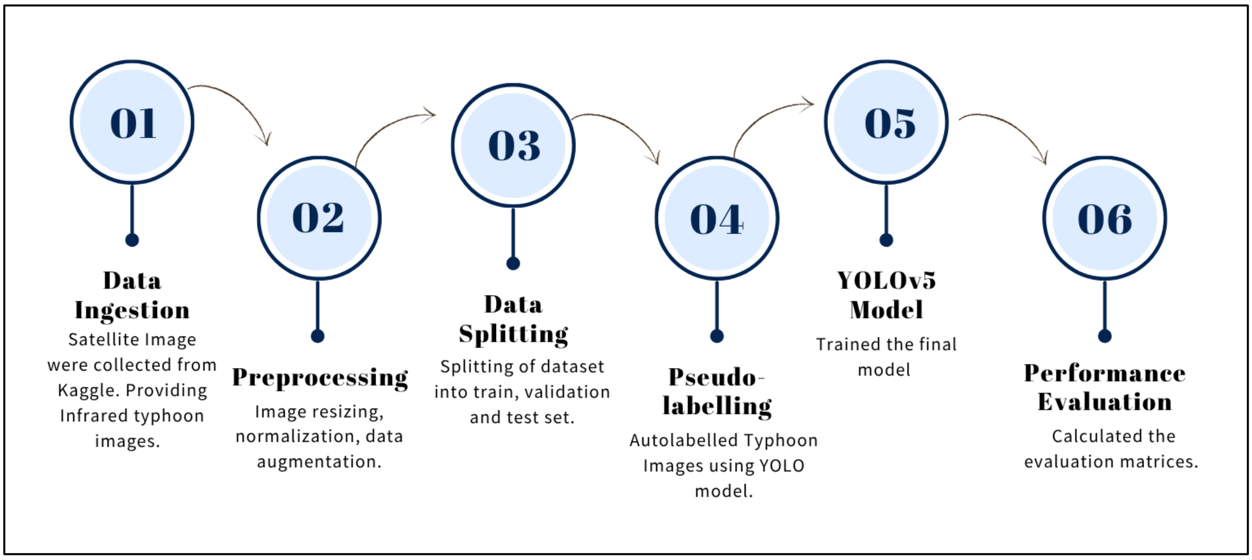


Figure 4: Project Workflow

4.1. Dataset description

The dataset used in this study consists of infrared satellite imagery specifically focused on typhoon systems. These images were sourced from the Himawari-8 satellite, operated by the Japan Meteorological Agency (JMA), which offers high-frequency and high-resolution observations ideal for real-time atmospheric monitoring. By taking data every ten minutes, Himawari-8 makes it possible to track meteorological events like tropical cyclones in great detail over time.

The imagery highlights cloud structures and thermal patterns associated with different stages of typhoon evolution, including genesis, intensification, peak strength, and dissipation. These stages are crucial for understanding the morphological progression of cyclones and for training machine learning models to recognize key features indicative of typhoon development.

The dataset was imported from Kaggle and consists around 70,000 images. Each satellite image in the dataset has a spatial resolution of 512 × 512 pixels, offering sufficient detail to capture core structural elements such as the eye, eyewall, and spiral rainbands. The dataset spans a diverse set of typhoons across different seasons and intensity levels, providing a balanced distribution of samples for robust training. This diversity helps in capturing a wide range of visual and thermal characteristics, improving the model's capability to localize and analyse typhoons under varying atmospheric conditions.

4.2. Image Preprocessing

4.2.1. Image Resizing and Normalization

```python
# Importing Libraries
import h5py
import os
from pathlib import Path
import matplotlib.pyplot as plt
import numpy as np
from PIL import Image
import cv2


base_directory = os.path.join(os.getcwd(), "typhon_images", "image")
h5_file_paths = []

# Target size for resizing
desired_size = (300, 300)

# collecting the .h5 files
for i in range(197901, 197902):
    target_path = os.path.join(base_directory, str(i))
    FILE_PATH = Path(target_path)          # converting the target path into a path object
    for file in FILE_PATH.glob("*.h5"):
        h5_file_paths.append(file)

print(h5_file_paths)

# opening and inspecting each image file
for path in h5_file_paths:
    with h5py.File(path, "r") as f:
        print(f"File: {path.name}")
        print("Keys in file:", list(f.keys()))
        infrared_data = f['Infrared']
        print(type(infrared_data))

```

(A)

```python
        #  checking shape and dtype of the dataset
        if isinstance(infrared_data, h5py.Dataset):
            print("Shape:", infrared_data.shape)
            print("Data type:", infrared_data.dtype)

            data = infrared_data[:]
            print(data)

    # First normalize to 0-255 for resizing if dtype is float
    data_min = np.min(data)
    data_max = np.max(data)
    data_scaled = 255 * (data - data_min) / (data_max - data_min)
    data_scaled = data_scaled.astype(np.uint8)

    # Resize using cv2 (height, width order)
    resized_array = cv2.resize(data_scaled, desired_size, interpolation=cv2.INTER_CUBIC)
    # Convert back to float normalized 0-1
    normalized = resized_array.astype(np.float32) / 255.0            # Image Normalization


    #  Directory for the normalized level
    normalized_output_dir = os.path.join(os.getcwd(), "typhon_images", "processed", "normalized")
    os.makedirs(normalized_output_dir, exist_ok = True)

    # Creating filename for saving the images
    filename = os.path.splitext(path.name)[0] + "_normalized.npy"
    normalized_img_path = os.path.join(normalized_output_dir, filename)

    # Saving the file
    np.save(normalized_img_path, normalized)
```

(B)

Figure 4.2.1: Code for Data Resizing and Normalization

Image processing is an important step in preparing satellite data for deep learning-based typhoon localization. Improving consistency and quality of input data, makes model training more successful. In order to guarantee that every satellite image had a consistent size and shape, a crucial component for batch processing in neural networks, OpenCV was utilized for image resizing. Using cv2.resize, each image was scaled to a specific dimension of 300 X 300 while preserving the general typhoon pattern structure. Following resizing, the pixel intensity values were scaled to a conventional range, usually between 0 and 1, using normalization. the speed of the learning process was stabilized by dividing the pixel value by 255.

4.2.2. Data Augmentation

```python
# Importing Libraries
import h5py
import os
from pathlib import Path
import matplotlib.pyplot as plt
import numpy as np
from PIL import Image
import cv2
import albumentations as A
from glob import glob


# Paths
image_dir = os.path.join(os.getcwd(), "label_images_png")
label_dir = os.path.join(os.getcwd(), "xml_label_images")
output_image_dir = "augmented_images"
output_label_dir = "augmented_labels"

# Create output folders if they don't exist
os.makedirs(output_image_dir, exist_ok=True)
os.makedirs(output_label_dir, exist_ok=True)

# Albumentations transform (you can customize)
transform = A.Compose([
    A.HorizontalFlip(p=0.5),
    A.VerticalFlip(p=0.5),
    A.RandomBrightnessContrast(p=0.2),
    A.ShiftScaleRotate(shift_limit=0.05, scale_limit=0.1, rotate_limit=15, p=0.7),
    A.RandomResizedCrop(size=(300, 300), scale=(0.8, 1.0), ratio=(0.75, 1.33), p=0.5)
], bbox_params=A.BboxParams(format='yolo', label_fields=['category_ids']))

```

(A)

```python
# Process each image
for img_path in glob(f"{image_dir}/*.png"):
    file_name = os.path.basename(img_path).split('.')[0]
    label_path = os.path.join(label_dir, file_name + ".txt")

    if not os.path.exists(label_path):
        print(f"[!] Label not found for {file_name}, skipping.")
        continue

    # Load image
    image = cv2.imread(img_path)
    height, width = image.shape[:2]

    # Load labels
    with open(label_path, 'r') as f:
        lines = f.readlines()

    bboxes = []
    category_ids = []
    for line in lines:
        parts = line.strip().split()
        if len(parts) != 5:
            continue
        cls, x, y, w, h = map(float, parts)
        bboxes.append([x, y, w, h])
        category_ids.append(int(cls))

```

(B)

```python
    # Perform 10 augmentations per image
    for i in range(10):
        try:
            augmented = transform(image=image, bboxes=bboxes, category_ids=category_ids)
            aug_img = augmented['image']
            aug_bboxes = augmented['bboxes']
            aug_ids = augmented['category_ids']

            # Save image
            aug_img_name = f"{file_name}_aug{i}.png"
            aug_img_path = os.path.join(output_image_dir, aug_img_name)
            cv2.imwrite(aug_img_path, aug_img)

            # Save updated label
            aug_label_path = os.path.join(output_label_dir, f"{file_name}_aug{i}.txt")
            with open(aug_label_path, 'w') as f:
                for bbox, cls in zip(aug_bboxes, aug_ids):
                    x, y, w, h = bbox
                    f.write(f"{cls} {x:.6f} {y:.6f} {w:.6f} {h:.6f}\n")
        except Exception as e:
            print(f"[!] Skipped {file_name}_aug{i} due to error: {e}")

print("Data augmentation completed! Check 'augmented_images' and 'augmented_labels'")
```

(C)

Figure 4.2.2: Code for Data Augmentation

Once these basic preprocessing steps were completed, a subset of (300) images were manually labelled by marking the centre of the typhoon in each image. These annotated images served as the foundation for supervised learning and were essential for training the initial model. After manual labelling, data augmentation was performed on this labelled subset as the final stage of preprocessing. Techniques such as rotation, flipping, zooming, brightness adjustments, and slight translations were applied to generate multiple variations of each image, while ensuring that the label coordinates were appropriately adjusted to match the transformed image. This significantly increased the diversity and size of the training dataset, making the model more robust to variations in real-world satellite imagery.

The model trained on the augmented, manually labelled images was then used to automatically label the remaining ~70,000 images in the dataset. This semi-automated labelling approach dramatically reduced the manual workload while maintaining consistent annotation quality across the entire dataset, enabling large-scale training for typhoon centre localization.

4.3 Data Splitting:

For the initial model training, a total of 3,000 augmented satellite images were systematically divided into training and validation sets to ensure balanced model evaluation. Approximately 80% of the data (2,400 images) was used for training, allowing the model to learn features related to typhoon centers across varied conditions. To prevent overfitting, about 20% (600 photos) of the data were set aside for validation, which was utilized to adjust hyperparameters and track performance during training.

This stratified data split ensured that the model's performance was fairly assessed and that it could effectively learn patterns while maintaining robustness across different image sets.

4.4. Initial Model Training

The initial model training began by splitting the dataset into training and validation sets in an 80:20 ratio to ensure balanced learning and evaluation. The YOLOv5 architecture was imported directly from the official GitHub repository, followed by the installation of required dependencies, including PyTorch and other supporting libraries. A custom data configuration yaml file was created to define the paths for training and validation images along with the class labels. With the setup complete, model training was initiated, leveraging YOLOv5's efficient object detection capabilities.

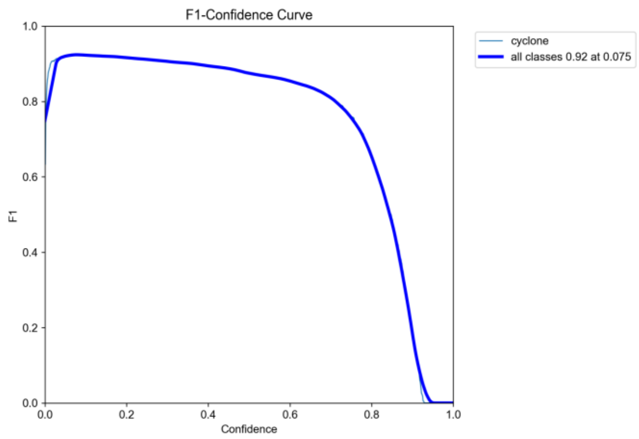
4.5. Final Model Training

The final model training was performed using the complete labelled dataset, which included both the manually annotated and the automatically labelled satellite images. The YOLO (You Only Look Once) architecture was used for its efficiency in real-time object localization. The dataset was divided into training, validation, and testing sets to ensure proper evaluation of the model's performance. Bounding box regression was used to train the YOLO model to predict the locations of typhoon centres in infrared satellite pictures. The model performance was increased by selecting the suitable hyperparameters like epochs, batch size and more. Accuracy and localization precision were evaluated using validation measures, and the model's loss was tracked to avoid overfitting. After training, the final model demonstrated strong generalization capabilities and was able to accurately detect typhoon centres in unseen satellite images, making it suitable for deployment in real-world storm monitoring applications.

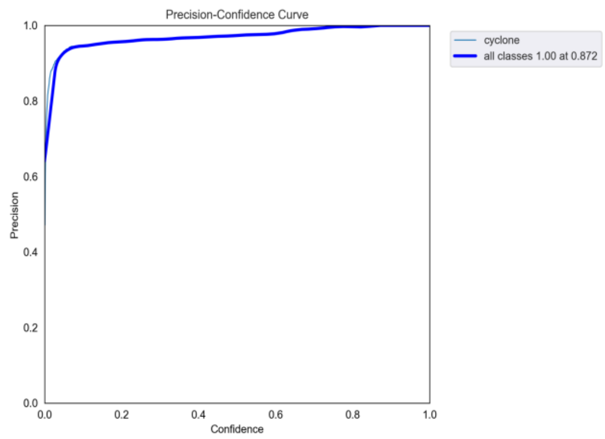
CHAPTER 5

RESULTS

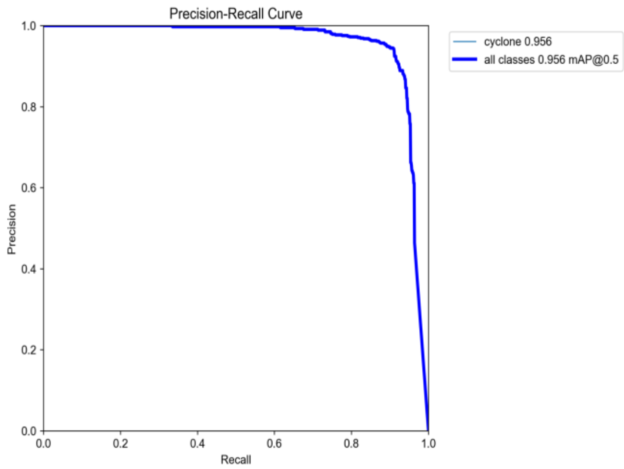
The YOLOv5-based typhoon centre localization model demonstrates the following performance.



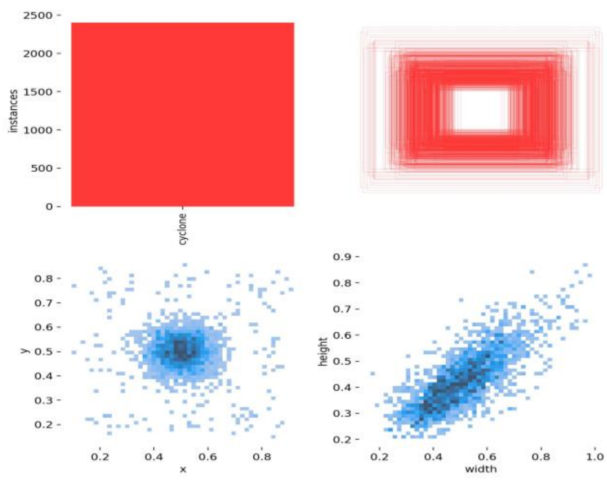
(A)



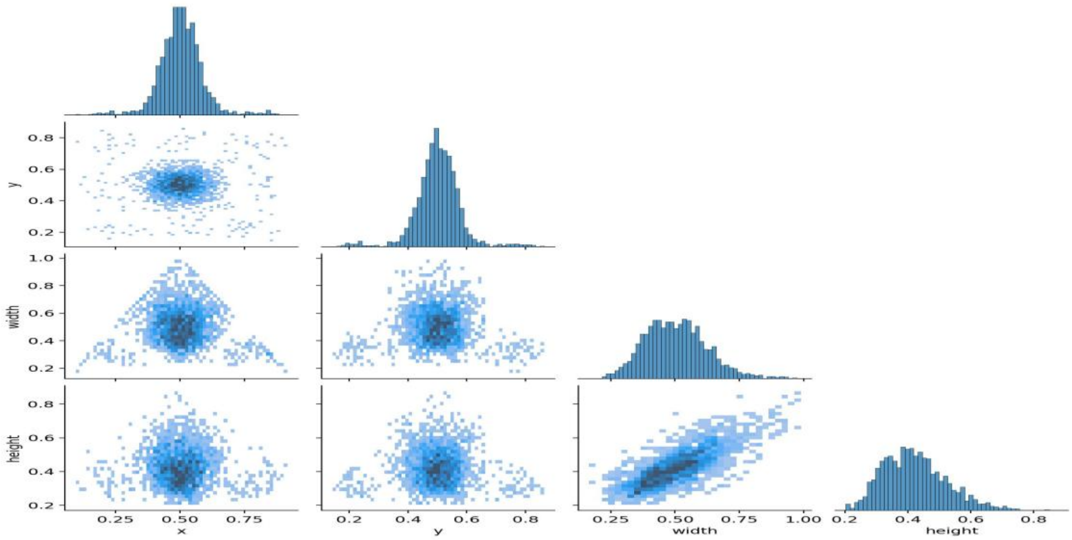
(B)



(C)



(D)



(E)

Figure 5: Performance Metrices

- **F1-Confidence Curve** (Figure. A): The model achieved a peak F1 score of **0.92** at a confidence threshold of **0.075**, indicating a good balance between precision and recall at lower confidence levels.
- **Precision-Confidence Curve** (Figure. B): Precision improves steadily with confidence, reaching **1.00** at **0.872**, reflecting high reliability in high-confidence predictions.
- **Precision-Recall Curve** (Figure. C): The model showed strong generalization, maintaining high precision throughout the majority of recall levels, and achieving a mAP@0.5 of 0.956.
- **Label Distribution** (Figure. D): Most cyclone centres are concentrated near the image centre (x, y ≈ 0.5), with bounding boxes showing strong correlation between width and height—consistent with typical storm morphology.
- **Bounding Box Statistics** (Figure. E): The distributions of x, y, width, and height are tightly clustered and normally distributed, indicating label consistency and model stability.

5.1. Performance Metrics

In this study on typhoon centre detection using satellite imagery, key performance metrics such as mean average precision (mAP) and the F1 score are used to evaluate the effectiveness of the YOLOv5 model. mAP is the primary metric, as it provides a comprehensive measure of both object localization and classification accuracy across all classes, making it highly suitable for complex remote sensing tasks like typhoon localization. Additionally, due to the inherent class imbalance in satellite datasets where typhoon centres occupy a small region compared to the overall image the F1 score is also considered. The F1 score offers a balanced view by combining precision and recall, effectively capturing the model's ability to correctly detect typhoon centres while minimizing both false positives and false negatives. Accuracy is deliberately excluded, as it can be misleading in imbalanced datasets, where most pixels may represent background rather than storm features [11].

1. **Precision**

   In the context of YOLOv5, precision refers to the proportion of correctly identified objects (true positives) out of all objects the model has predicted. A high precision value means that most of YOLOv5's detections are accurate.

$$\text{Precision} = \frac{TP}{(TP + FP)}$$

2. **Recall (Sensitivity)**

Recall measures YOLOv5's effectiveness in detecting all actual objects present in the dataset. It represents the ratio of true positive detections to all necessary items (true positives plus false negatives). Fewer missed detections are indicated by a higher recall.

$$\text{Recall} = \frac{TP}{(TP + FN)}$$

3. **Score**

The F1 score balances precision and recall by calculating their harmonic mean. In datasets with class imbalance, like in this study, F1 is a more reliable metric than accuracy. It provides a single score that reflects how well YOLOv5 avoids both false positives and false negatives.

$$\text{F1 Score} = 2 \times \frac{(\Pr ecision \times Recall)}{(\Pr ecision + Recall)}$$

4. **mAP (Mean Average Precision)**

mAP is the primary metric used to evaluate YOLOv5's performance in this study. It represents the average precision across all object classes and Intersection-over-Union (IoU) thresholds. mAP reflects both the accuracy of object localization and classification, making it a standard metric in object detection tasks.

mAP = Average of APs across all IoU thresholds and classes

CHAPTER 6

LIMITATIONS

While the proposed deep learning-based approach for typhoon centre localization shows promising results, several limitations exist that must be acknowledged to understand the scope and boundaries of this study.

1. Computational Resource Dependence: Although YOLOv5 is optimized compared to earlier versions, training and inference especially with larger variants like YOLOv5x still demand high-end GPUs and substantial memory. This dependency can be a significant barrier when deploying the model in real-time or in resource-constrained environments such as remote weather stations or edge devices.
2. Complexity of Model Deployment: Deploying a YOLO-based model in production settings is not always straightforward. Integration with mobile platforms, embedded systems, or cloud-based infrastructure often requires model conversion (e.g., to ONNX or TensorRT), which involves additional technical expertise and tuning for performance optimization.
3. Trade-offs Between Speed and Accuracy: YOLOv5 offers various model sizes (nano, small, medium, large, extra-large), each presenting trade-offs between inference speed and detection accuracy. While lighter models provide faster processing, they may sacrifice precision—particularly for small or ambiguous typhoon signatures. Choosing the optimal model variant for a specific use case requires careful balancing.
4. Model Bias from Initial Manual Labelling: The model was initially trained on a manually labelled subset of 300 satellite images. Since these annotations served as the foundation for automatic labelling of the remaining ~70,000 images, any errors or inconsistencies in the initial labels may propagate through the entire dataset. This introduces potential bias in model learning, as the early labels may not represent the full diversity of typhoon appearances.
5. Generalizability to Other Satellite Sources: The dataset was sourced exclusively from Himawari-8 infrared imagery. While effective for this specific satellite and region, the model's performance may degrade when applied to images from different satellites (e.g., GOES, Meteosat) or regions with different atmospheric characteristics. Additional fine-tuning would likely be necessary to ensure cross-platform generalization.
6. Limited Temporal Coverage: Despite Himawari-8's high frequency of image capture (every 10 minutes), certain temporal limitations exist. Differences in illumination between day and night images or cloud coverage variations can affect feature visibility. These inconsistencies may

reduce the model's ability to generalize across all timeframes, especially if night-time imagery is underrepresented during training.

7. Label Noise from Auto-labelling: The use of a semi-automated process to label over 70,000 images introduces the risk of label noise. While the initial model facilitated large-scale annotation, its predictions may not always be accurate, particularly in edge cases such as weak or disorganized storms. These noisy labels can negatively impact model performance during final training stages.

Considering these limitations, further work is required to refine the model for broader operational use. This includes collecting more diverse data, incorporating validation checks during auto-labelling, and testing the model on multiple satellite sources and typhoon conditions.

CHAPTER 7

ADVANTAGES

1. High Speed and Real-Time Processing: One of the biggest strengths of YOLO is its ability to detect objects almost instantly. Since it looks at the entire image in one go, it drastically cuts down processing time. This is especially useful for tracking fast-moving weather systems like typhoons, where every minute counts. The model's speed makes it ideal for real-time applications where rapid decision-making is essential.
2. Improved Accuracy: With each version, YOLO has gotten better at accurately identifying objects even when they're small, unclear, or surrounded by noise. In the case of typhoons, which often appear as large but diffuse cloud systems, YOLOv5 does a good job of pinpointing their centres. This accuracy can be critical for understanding storm paths and intensities.
3. Versatility and Model Variant Options: YOLOv5 comes in different sizes from lightweight versions to more powerful ones. This flexibility means the model can be used on various platforms. For example, smaller versions can run on low-power devices near the observation site, while larger models can be deployed on cloud servers for more detailed analysis.
4. Efficient Architecture: The overall design of YOLOv5 is both clean and practical. It's built to make the most of your hardware and train quickly without sacrificing performance. Thanks to PyTorch and the use of advanced components like CSPNet, it's also easier for developers to understand, tweak, and integrate into different systems.
5. Robust Data Augmentation: YOLOv5 includes several smart data augmentation techniques like image mixing, colour adjustment, Mosaic, MixUp, HSV (Hue, Saturation, Value) adjustments and random transformations that help the model get better at handling real-world images. These tricks make the model more resilient to things like lighting changes, cloud patterns, and other unpredictable variations in satellite imagery.
6. Application-Specific Benefits: This model isn't just about good performance it can make a difference. By quickly and accurately identifying typhoon centres, it helps meteorologists and emergency teams act faster. Quicker warnings mean more time for communities to prepare, potentially saving lives and reducing damage. It also reduces the manual workload for forecasters, helping them focus on critical decision-making.
7. Integration Potential with Early Warning Systems: The automated nature of the model makes it a strong candidate for integration with national and regional alert systems. For example, platforms used by the Indian Meteorological Department (IMD) or the National Disaster Management Authority (NDMA) could use this model to automatically trigger early alerts. This

kind of integration could strengthen our response to severe weather events and improve overall public safety.

CHAPTER 8

CONCLUSION

This study demonstrates the potential of deep learning, particularly the YOLOv5 architecture, in addressing the challenges of object detection within satellite-based typhoon imagery. The model effectively tackled complexities such as varying cloud densities, dynamic storm patterns, and limited labelled data. Through systematic training, careful validation, and metric-based performance analysis, YOLOv5 exhibited strong capabilities in accurately localizing typhoon centres, even under conditions of class imbalance and visual clutter. The use of robust performance metrics mean average precision (mAP) and F1 score provided reliable insights into the model's detection accuracy and consistency across diverse samples. These results highlight YOLOv5's suitability for high-resolution meteorological image analysis and confirm its potential for real-world deployment. With continued refinement, including domain-specific tuning and integration with automated alert systems, such models can significantly enhance early warning mechanisms and support timely, data-driven decision-making in meteorology and disaster preparedness.

REFERENCES

[1] P. Nandal, P. Mann, N. Bohra, G. Aldehim, A. A. H. Elnour, and R. Allafi, “Tropical cyclone intensity estimation based on YOLO-NAS using satellite images in real time,” *Alexandria Engineering Journal*, vol. 113, pp. 227–241, Feb. 2025, doi: 10.1016/j.aej.2024.10.072.

[2] C. Harinakshi, K. Sigin, S. Sreenivas, P. Roshan Shetty, and H. Mohammed Rashiq, “Deep Learning Model for Estimating Predicting and Forecasting of Tropical Cyclone Intensity using Satellite Images,” in *2023 IEEE International Conference on Distributed Computing, VLSI, Electrical Circuits and Robotics, DISCOVER 2023*, Institute of Electrical and Electronics Engineers Inc., 2023, pp. 281–286. doi: 10.1109/DISCOVER58830.2023.10316711.

[3] P. Jiang, D. Ergu, F. Liu, Y. Cai, and B. Ma, “A Review of Yolo Algorithm Developments,” in *Procedia Computer Science*, Elsevier B.V., 2021, pp. 1066–1073. doi: 10.1016/j.procs.2022.01.135.

[4] S. Jiang, H. Huang, J. Yang, X. Zhang, and S. Wang, “Innovative Research on Small Object Detection and Recognition in Remote Sensing Images Using YOLOv5,” in *International Archives of the Photogrammetry, Remote Sensing and Spatial Information Sciences - ISPRS Archives*, International Society for Photogrammetry and Remote Sensing, May 2024, pp. 77–83. doi: 10.5194/isprs-archives-XLVIII-4-W10-2024-77-2024.

[5] H. Sakaino, N. Gaviphatt, A. Insisiengmay, L. Zamora, D. F. Ningrum, and S. Kotsuki, “DeepTrCy: Life Stage Identification of Satellite Tropical Cyclone Images,” in *International Geoscience and Remote Sensing Symposium (IGARSS)*, Institute of Electrical and Electronics Engineers Inc., 2023, pp. 5186–5189. doi: 10.1109/IGARSS52108.2023.10282363.

[6] K. PushpaRani, G. Roja, R. Anusha, C. Dastagiraiah, B. Srilatha, and B. Manjusha, “Geological Information Extraction from Satellite Imagery Using Deep Learning,” in *2024 15th International Conference on Computing Communication and Networking Technologies (ICCCNT)*, IEEE, Jun. 2024, pp. 1–7. doi: 10.1109/ICCCNT61001.2024.10724802.

[7] L. He, L. Xiao, P. Yang, and S. Li, “Typhoon localization detection algorithm based on TGE-YOLO,” *Sci Rep*, vol. 15, no. 1, p. 3385, Jan. 2025, doi: 10.1038/s41598-025-87833-8.

[8] S. K. Jaiswal and R. Agrawal, “A Comprehensive Review of YOLOv5: Advances in Real-Time Object Detection,” *International Journal of Innovative Research in Computer Science and Technology*, vol. 12, no. 3, pp. 75–80, May 2024, doi: 10.55524/ijircst.2024.12.3.12.

[9] D. Jayakumar and S. Peddakrishna, “Performance Evaluation of YOLOv5-based Custom Object Detection Model for Campus-Specific Scenario,” *International Journal of Experimental Research and Review*, vol. 38, pp. 46–60, Apr. 2024, doi: 10.52756/ijerr.2024.v38.005.

[10] M. Wang *et al.*, “Research on Typhoon Multi-Stage Cloud Characteristics Based on Deep Learning,” *Atmosphere (Basel)*, vol. 14, Dec. 2023, doi: 10.3390/atmos14121820.

[11] M. A. Kenk and M. Hassaballah, “DAWN: Vehicle Detection in Adverse Weather Nature Dataset,” Aug. 2020, doi: 10.17632/766ygrbt8y.3.